MILITARY SPECIFICATION

MICROPHONE, CARBON, NOISE CANCELLING, TYPE M-58/U

1. SCOPE

1.1 This specification covers a handheld, noise cancelling, carbon microphone unit capable of transmitting voice, while discriminating against ambient noise when operated in conjunction with Naval communication transmitters and associated equipment.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawing, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-571 - Solder, Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.
QQ-Z-325 - Zinc Coating, Electrodeposited; Requirements for.

MILITARY

MIL-C-15328 - Coating, Pretreatment (Formula No. 117 for Metals).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

BUREAU OF SHIPS

RE51C125 - Microphone, M-58/U.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

B-103-55 - Phosphor Bronze, Plate, Sheet, Strip and Roll Bar.
B-139-58 - Phosphor Bronze, Bars, Rods, and Shapes.
B-194-55 - Copper-Beryllium Alloy, Plate, Sheet, Strip, Roll, Bar

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Preproduction sample. - Two preproduction samples shall be furnished for the tests specified in 4.3.

3.2 Material - The materials used shall be as hereinafter specified. Materials not specified shall be of the quality best suited for the purpose intended.

3.2.1 Molded plastic material. - All molded plastic parts shall be of the high temperature, high impact resistant plastic material which shall meet the environmental requirements of this specification.

3.2.2 Steel parts. - Steel parts shall be cadmium plated in accordance with class 1, type II of Specification QQ-P-416, with supplementary chromate treatment, or zinc-plated in accordance with type II, class 2 in accordance with QQ-Z-325.

3.2.3 Exposed non-ferrous parts. - Parts that are exposed after the microphone is assembled shall be thoroughly cleaned, followed by a preventive coating in accordance with Specification MIL-C-15328. This shall be followed by two coats of a high quality alkyd, semi-gloss, black enamel.

3.2.4 Contacts. - All metal parts of the microphone unit which make electrical contact with the switch in the handgrip shall be made of coin silver, or shall be flash-plated with nickel and then gold-plated. Contact shall be effected by spring pressure.

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All electrical contacts of the switch mechanism shall be coin silver.

3.2.5 Metal switch parts. - Metal switch parts, except contacts, shall be phosphor bronze in accordance with ASTM designation B-103-55 or B-139-58, or suitable tempered beryllium copper in accordance with ASTM designation B-194-55.

3.2.6 Carbon granules. - Carbon granules shall be of a quality, size, and shape suitable in every respect for this particular application. The carbon granules shall be retained positively in the microphone unit in such a manner that it is impossible for them to leak from the carbon cavity during service use of under any of the conditions specified herein. All surfaces of the carbon chamber which are in contact with the carbon granules shall be suitably plated with gold. The plated surface shall be free from porosity and thoroughly cleaned before assembly.

3.2.7 Cementing compound. - Any cementing compound used in the construction of the microphone shall be of such a character and quality that it will withstand all service and test conditions specified herein without evidence of loosening or otherwise affecting the performance requirements of the microphone.

3.2.8 Soldering flux. - Only resin or resin solder may be used as a flux in the assembly of the microphone, or any part thereof.

3.2.9 Soft solder. - Soft solder shall not be depended upon for the mechanical strength of any soldered joint. There shall be no evidence of "cold soldering", and the use of excessive amounts of solder will not be acceptable. Soft solder shall be solid or rosin flux-cored in accordance with composition Sn60 of Specification QQ-S-571.

3.3 Detail requirements. -

3.3.1 Design. - The microphone units shall be of the noise-cancelling, carbon type. In general, the design and assembly of the microphone units shall be such that they will be inherently stable in mechanical, electrical, and acoustical characteristics. It shall not be necessary to condition a microphone unit by any special procedure before it is ready for use.

3.3.2 Dimensions. - The outer dimensions of the microphone units shall be in accordance with drawing RE51C125.

3.3.3 Microphone circuit. - The microphone circuit shall be as specified in Drawing RE51C125.

3.3.3.1 Cord terminations. - One end shall be terminated in a plug. The other end shall be removable from the microphone by disconnecting the screw type terminals therein, and by removing the seal in the housing. No special tools shall be required. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the office of the Government inspector.)

3.3.3.2 Cord anchorage. - The cord shall be anchored at the microphone and plug to withstand a tension not less than ten pounds.

3.3.3.3 Strain reliever. - Means shall be provided whereby tension applied to the cord is not transmitted to the terminals. A strain reliever shall protect the cord from breaking due to cord flexure at the junction of the cord and housing.

3.3.3.4 Terminals. - The terminal inserts shall be suitably molded into the housing or terminal block. Wires, where required, shall be connected to the terminals with suitable lugs. Other connections shall be by means of contacts, such that the transmitter unit may be disassembled from the housing without unsoldering any connections. Contacts shall be of suitably hardened beryllium copper, silver-plated or gold-plated.

3.3.4 Switch. - The "press-to-talk" switch shall be such that it may be easily actuated with the operator wearing gloves. The switch shall be of the spring return type, or equal. The switch shall be of the momentary contact type, which shall close the carbon button circuit, and an external relay circuit to ground while the button is held depressed. It shall be assembled in the housing in a thoroughly watertight manner.

3.3.4.1 Switch action. - All contacts in the switch mechanism shall "make" without chattering in any position of the microphone. Closing the switch contacts for relay and microphone operation shall occur approximately simultaneously. The switch of the completely assembled microphone shall satisfactorily withstand the switch action test specified in 4.6.4.

3.4 Performance. -

3.4.1 Direct current (d.c.) resistance. - The d.c. resistance of the microphone shall be within 100 ohms $\pm$ 50 ohms when tested as specified in 4.6.7.

3.4.2 Frequency response. - In a vertical position the output at 1000 cycles per second (c.p.s.) across a noninductive load of 100 ohms (see 4.6.8) shall be between plus 46 and plus 53 decibels (db) (re 1 millivolt) when a sound pressure level of 115 db relative to 0.0002 dynes per square centimeter is applied to the microphone. The response between 300 to 1000 c.p.s. shall be within plus or minus 3 db of the output at 1000 c.p.s.,

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and between 1000 and 4000 c.p.s. shall be within plus or minus 6 db of the output at 1000 c.p.s. The response below 300 c.p.s. and above 4000 c.p.s. shall fall as rapidly as practicable.

3.4.2.1 Positional response. - The response shall not fall below plus 40 db over the range 300 to 4000 c.p.s., except in the minus 90 and minus 45-degree positions shown in table I. Zero degree in table I indicates microphone "working face" in the vertical plane. Positive angles indicate "face" upwards. Negative angles indicate "face" down.

3.4.3 Noise discrimination. - The microphone shall withstand the noise discrimination test of 4.6.9.

Table I - Positional response.

Frequency range (c.p.s.)	Deviation in db from 1000 c.p.s. value at the position indicated			
	Plus 90 degrees	Plus 45 degrees	Minus 45 degrees	Minus 90 degrees
300 to 1000	Plus 3 Minus 5	Plus 3 Minus 5	Plus 3 Minus 5	Plus 3 Minus 5
1000	0 0	0 0	0 0	0 0
1000 to 3000	Plus 8 Minus 6	Plus 8 Minus 6	Plus 8 Minus 6	Plus 12 Minus 6
3000	Plus 8 Minus 6	Plus 8 Minus 6	Plus 8 Minus 6	Plus 8 Minus 6
3000 to 4000	Plus 9 Minus 6	Plus 6 Minus 6	Plus 6 Minus 10	Plus 6 Minus 15

3.4.4 Current rating. - The nominal rated button current shall be 60 milliamperes.

3.4.4.1 Anti-packing. - The microphone shall not "pack" when subjected to an excitation of current up to 80 milliamperes for a minimum of 6 hours.

3.4.4.2 Burning noise. - The output of the microphone unit due to burning noise shall be kept to the lowest practicable minimum, and in no case shall it exceed 1.0 millivolt when tested as specified in 4.6.11.

3.4.5 Response variation after test. - After being subjected to each and all of the tests, the response of the microphone over the frequency range of 300 to 4000 c.p.s. shall not vary more than the following:

(a) Plus or minus 3 db from table I values, after tests as follows:
  (1) Temperature cycling (see 4.6.5).
  (2) Temperature extremes (see 4.6.6).
  (3) Impact (see 4.6.2).
  (4) Antipacking (see 4.6.12).
  (5) Altitude when specified (see 4.6.3 and 6.1).

(b) Plus or minus 8 db from table I values after the salt spray test specified in 4.6.1.

3.4.6 Salt spray. - After being subjected to the salt spray test specified in 4.6.1, there shall not be any signs of peeling or flaking of the microphone case or to the finish of the metallic parts. There shall also not be any sign of moisture entering the microphone.

3.4.7 Insulation resistance. - The insulation resistance shall be not less than 100 megohms as measured with a 500-volt source, when tested in accordance with 4.6.13.

3.4.8 Impact. - There shall be no signs of breakage of the microphone after it has been subjected to the impact test Specified in 4.6.2.

3.5 Weight. - The assembled microphone shall be as light as possible commensurate with the requirements specified herein.

3.6 Markings. - The nomenclature, manufacturer's prefix, and stock number shall be legibly marked in the approximate locations specified on Drawing RE51C125. All markings shall be permanent for the normal life of the microphone.

3.7 Workmanship. - Workmanship shall be such that the finished product shall meet all the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Test condition. - Unless otherwise specified herein, all measurements and tests shall be made at room ambient temperature, pressure, and humidity.

4.3 Preproduction inspection. -

4.3.1 A preproduction sample consisting of two microphones shall be made and assembled by the manufacturer with tools and methods that, as far as practicable, are the same as those which will be used for quantity production. The manufacture of microphones for Government contract prior to approval of the preproduction sample shall be at the manufacturer's risk.

4.3.2 Preproduction inspection shall consist of the examination and tests specified in table II.

Table II - Preproduction inspection.

Subject	Requirement paragraph	Reference
Visual examination	3.2.1.1, 3.3.2, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.4, 3.5, 3.6, 3.7	4.5
D.c. resistance	3.4.1	4.6.7
Frequency response	3.4.2	4.6.8
Noise discrimination	3.4.3	4.6.9
Positional response	3.4.2.1	4.6.10
Switch action	3.3.4.1	4.6.4
Altitude (when specified)	3.4.5	4.6.3
Burning noise	3.4.4.2	4.6.11
Anti-packing	3.4.4.1, 3.4.5	4.6.12
Temperature cycling	----	4.6.5
Temperature extremes	3.4.5	4.6.6
Salt spray	3.4.5, 3.4.6	4.6.1
Insulation resistance	3.4.7	4.6.13
Impact	3.4.5, 3.4.8	4.6.2

4.3.3 Failures and resubmissions. - Failure of any specimen in any one of the tests will be cause for refusal to grant approval. In the event of failure, the manufacturer may be requested to submit another sample equal in number to the original submission, accompanied with test data and a description of the changes made to correct the cause of failure.

4.4 Sampling for acceptance inspection. -

4.4.1 Inspection lot. - All microphones presented at one time shall be considered a lot for purposes of sampling and inspection.

4.4.2 Acceptance inspection. - Acceptance inspection shall consist of groups A and B examination and tests specified in 4.4.3 and 4.4.4. The samples shall be subjected to the tests in the order shown.

4.4.3 Group A examination and tests. - Group A examination and tests shall be as specified in table III. The sample shall be selected in accordance with inspection level II of Standard MIL-STD-105. The Acceptable Quality Level shall be 1.5 percent.

Table III - Group A inspection.

Subject	Requirement paragraph	Reference paragraph
Visual and dimensional examination	3.2.1.1, 3.3.2, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.4, 3.5, 3.6, 3.7	4.5
Burning noise	3.3.4.2	4.6.11
Direct current resistance	3.4.1	4.6.7

4.4.4 Group B tests. - Group B tests shall consist of the tests specified in table IV. The samples shall be selected in accordance with inspection level I of Standard MIL-STD-105. The Acceptable Quality Level shall be 4 percent.

Table IV - Group B tests.

Subject	Requirement paragraph	Reference paragraph
Anti-packing	3.3.4.1, 3.3.5	4.6.12
Frequency response	3.3.2	4.6.8
Positional response	3.3.2.1	4.6.10
Noise discrimination	3.3.3	4.6.9
Insulation resistance	3.3.7	4.6.13

4.5 Visual and dimensional examination. - Each of the sample microphones selected shall be examined to determine conformance with this specification with respect to finish, marking, workmanship, design, weight and dimensions.

4.6 Test procedures. -

4.6.1 Salt spray. - The salt spray test shall be conducted in accordance with condition "A" of Standard MIL-STD-202. The microphone shall then meet the requirements of 3.4.5 and 3.4.6.

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4.6.2 Impact. - The microphone shall be dropped ten times at random orientation from a height of 5 feet onto a hardwood floor. The microphone shall then meet the requirements specified in 3.4.5 and 3.4.8.

4.6.3 Altitude. - The microphone shall not be damaged by reduced pressure equivalent to that found at 50,000 feet elevation for 30 minutes. It shall provide intelligible communication under these conditions. Upon return to normal conditions, the unit shall be promptly tested for conformance to 3.4.5.

4.6.4 Switch action. - The complete switch assembly shall satisfactorily withstand, without electrical or mechanical failure, 100,000 complete make and break cycles at a rate of 25 cycles per minute while carrying a current of 1 ampere at 110 volts, 60 c.p.s.

4.6.5 Temperature cycling. - The microphone shall be subjected to five temperature cycles as follows:

(a) 30 minutes at minus 60°C. to 75° centigrade (C.).
(b) 10 minutes at room temperature.
(c) 30 minutes at plus 75° C. The transition time shall be minimum.

4.6.6 Temperature extremes. - Temperature extremes shall range from minus 60° $^{+5^\circ}_{-0^\circ}$ C. for 24 hours and plus 75° $^{+0^\circ}_{-5^\circ}$ C. for 24 hours. Tests are to be initiated within 1 minute after exposure. The microphone shall then meet the requirements of 3.4.5.

4.6.7 D.c. resistance. - The d.c. resistance of the microphone unit shall be measured with the rated button current applied, before the frequency response test of 4.6.8. The resistance shall be calculated from the voltage across the microphone and the button current, with a sound pressure level of 115 db at 1000 c.p.s. established at the face of the microphone. The voltage and current readings shall be taken 10 seconds after the application of the signal.

4.6.8 Frequency response. - Tests shall be conducted to determine conformance with 3.4.2.

4.6.9 Noise discrimination. - Noise discrimination shall be measured in accordance with the following method:

(a) By means of suitable apparatus similar to that detailed in figure 1, or equal, and the test circuit of figure 2, or equal, provision shall be made for the production of the noise spectrum, table V, at an r.m.s. sound pressure of 115 db above a reference level of 0.0002 dyne per square centimeter as measured at the microphone diaphragm. The "noise" source shall be a loud speaker capable of producing a sound pressure of 115 db above a reference level of 0.0002 dyne per square centimeter at the rear as well as the front of the microphone and the loud speaker shall be located as far away from the microphone as practicable. The calibrating microphone shall be mounted in the test rig, in the box, 1/4 inch from the mouth of the signal speaker which shall be short-circuited during adjustments of the noise spectrum.

Table V - Noise spectrum

Frequency (cps)	Sound pressure in db relative to sound pressure at 130 cps
40	0
70	0
130	0
300	-5
600	-9
1000	-13
2000	-17
3000	-19
4000	-21

(b) By means of the apparatus as set up previously, provision shall also be made for the production of the following signal spectrum, table VI, at a r.m.s. sound pressure of 115 db above a reference level of 0.0002 dyne per centimeter. The "signal" source shall be the driver unit. The calibrating microphone shall be positioned 1/4 inch directly in front of the driver unit.

Table VI - Signal spectrum.

Frequency (cps)	Sound pressure in db relative to sound pressure at 130 cps
130	0
300	+7
600	+8
1000	+5
2000	-3
3000	-7
4000	-9

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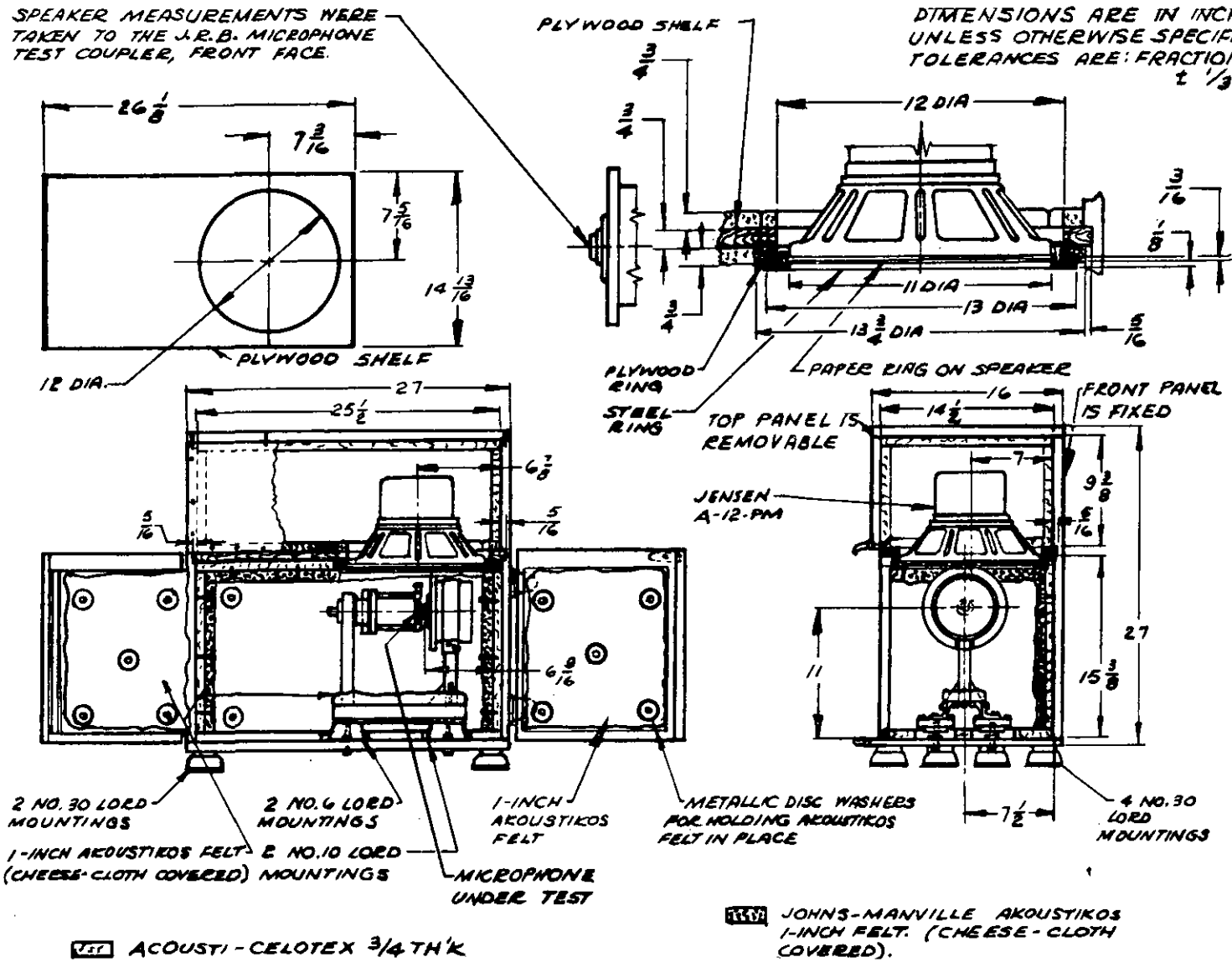


Figure 1 - Outline of test apparatus for noise discrimination.

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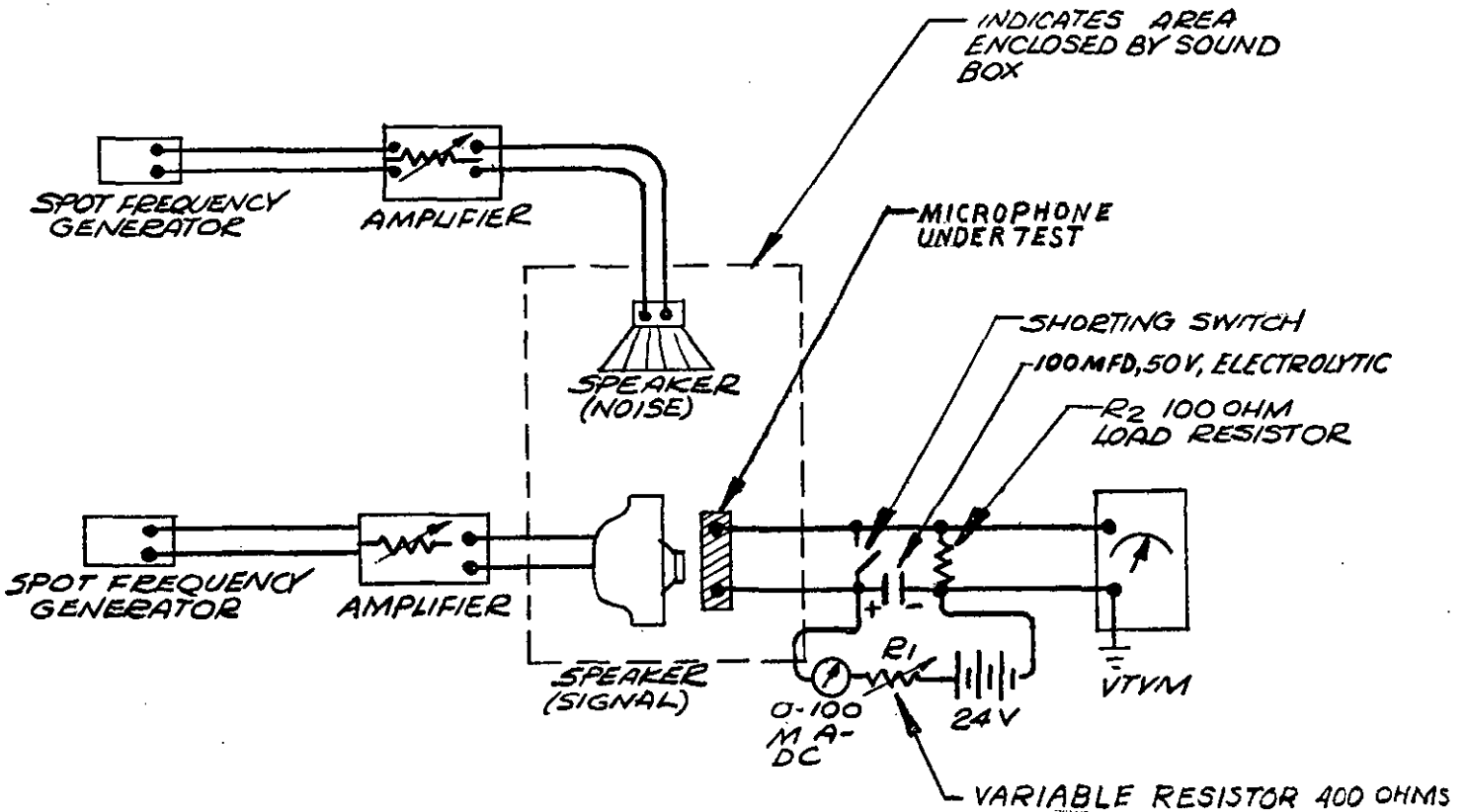


Figure 2 - Outline of test circuit for noise discrimination.

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(c) The microphone under test shall be mounted in its test rig with its front grid 1/4 inch from the mouth of the signal speaker. The microphone shall be connected to the test circuit.
(d) The short-circuit current shall be in accordance with 4.6.9(e). The short-circuit switch shall be opened and the noise spectrum applied for 3 seconds. The noise spectrum shall be removed, the signal spectrum applied for 3 seconds, and the output of the microphone shall be measured. The signal spectrum shall be removed, the noise spectrum applied for 3 seconds, and the output of the microphone measured. The signal-to-noise ratio shall be the quotient of these values.
(e) Switch S-1 of the test circuit (figure 2) shall be closed (no microphone current) and resistor R-1 shall be adjusted to produce a reading of 60 milliamperes on the microphone.

4.6.10 Positional response. - Test shall be conducted to determine conformance with 3.3.2.1.

4.6.11 Burning noise. - The output of the microphone unit due to burning noise shall be measured in the circuit shown on figure 3. After the microphone unit has been conditioned as specified in 4.6.9, it shall remain still and without signal input for 30 seconds. The maximum output, as indicated on meter M-2 of the test circuit during the 10-second interval immediately following the 30-second period, shall be the measure of the burning noise.

4.6.12 Anti-packing. - Anti-packing shall be made to determine conformance with 3.4.4.1.

4.6.13 Insulation resistance. - The transmitter cartridge shall be disconnected from the circuit before the insulation resistance is measured. It shall be measured between all circuit conductors and between all circuit conductors and metal parts of the microphone. The insulation resistance shall meet the requirements of 3.4.7.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early installation. -

5.1.1 Microphone. -

5.1.1.1 Preservation and packaging. - Microphones shall be preserved and packaged to provide protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Microphones, packaged as specified, shall be packed in a manner which will be acceptable by the common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers shall comply with the carrier's rules and regulations applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on containers in accordance with the contractor's method of application. In addition to any special marking required by the contract or order, the information shall include the Federal stock number, nomenclature, contract or purchase order number, contractor's name, and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment which may be required when procurement is made.

5.2.1.1 Preservation and packaging, packing, and marking. - Microphones shall be preserved and packaged by level A or C; packed by level A or B; and marked in accordance with Specification MIL-E-17555.)

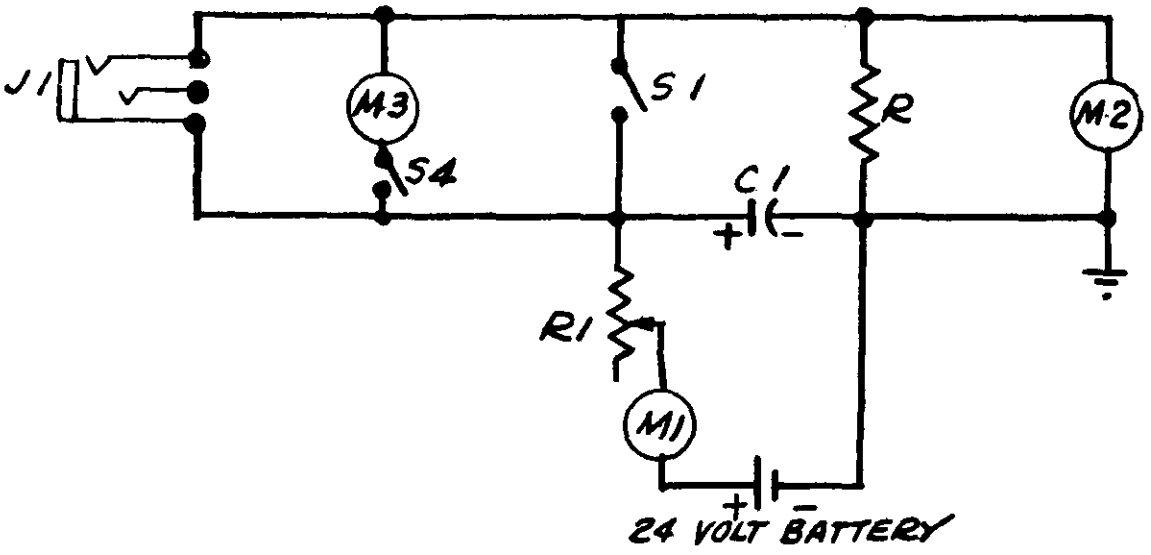
6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether altitude test is required (see 3.4.5 and table II)
(c) Preservation, packaging, packing, or marking requirements other than those required by 5.1 (see 5.2).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded

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J1 - MICROPHONE JACK, 3-CONTACT
S1 - SWITCH, SPST
C1 - CAPACITOR, 100 MFD, 50 VOLT, ELECTROLYTIC
R1 - VARIABLE RESISTOR, 400 OHM
R - RESISTOR, 100 OHM
M1 - MILLIAMMETER, 0-200 MA, DC.
M2 - ELECTRONIC VOLTMETER, BALLANTINE MODEL 300, OR EQUAL
M3 - VOLTMETER, 0-10 VOLTS, DC, 1000 OHMS PER VOLT, MINIMUM
S4 - SWITCH, PUSH BUTTON, SINGLE CONTACT

Figure 3 - Wiring diagram for burning noise test.

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Preparing activity:
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